INCH-POUND

MIL-S-24187/13(SH)
22 March 1991

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, STYLE VLLS TYPE 5VLLSO-1C

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

INCHES	MM	INCHES	MM
.03	0.76	3.060	77.72
.09	2.29	3.96	100.58
.12	3.05	9.40	238.76
.28	7.11	9.96	252.98
.312	7.93	14.00	355.60
.45	11.43	16.46	418.08
		16.67	423.42

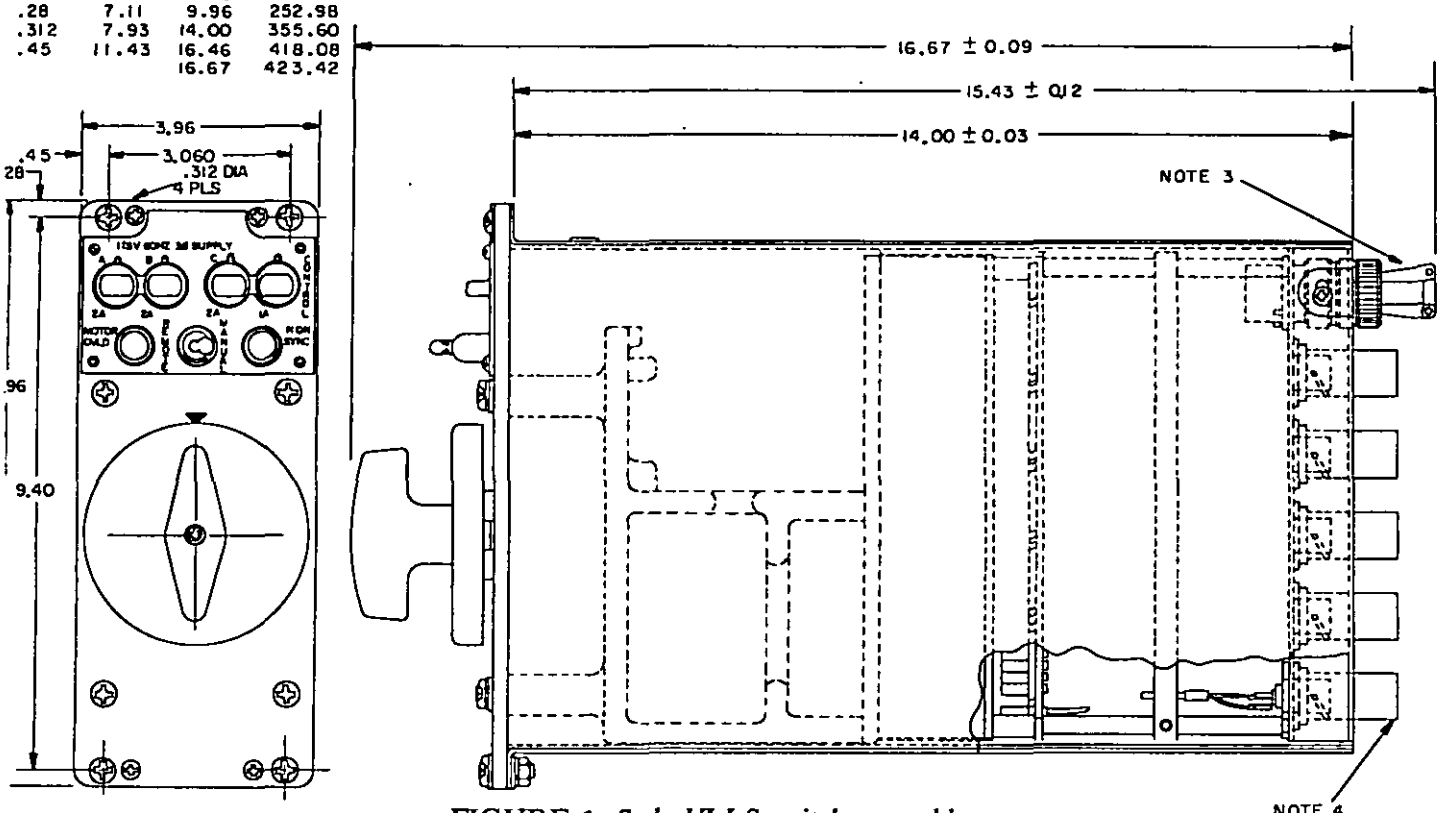


FIGURE 1. *Style VLLS switch assembly.*

AMSC N/A FSC 5930

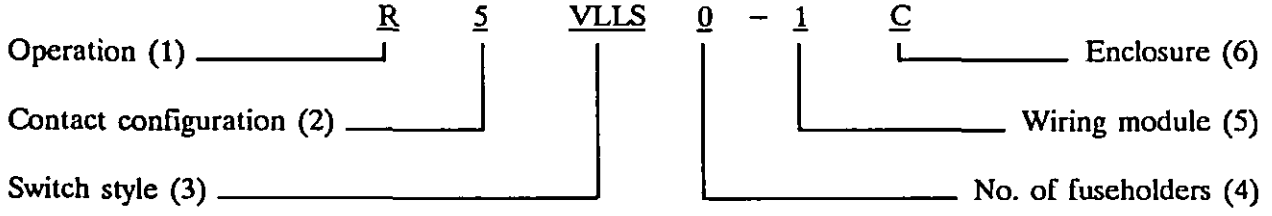
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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Remote switch, connector M28840/17AC1S1 (21 pin) and cover M28840/13 CAW, supplied only when specified.
4. Coax connector, P/N PL50N CAGE 14949, supplied only when specified.

FIGURE 1. *Style VLLS switch assembly – Continued.*

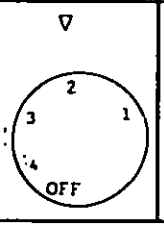
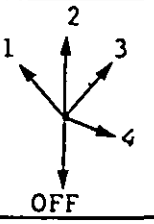
Metric equivalents (to the nearest 0.01mm) are given for general information only and are based upon 1 inch = 25.4mm



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.
(2) Contact configuration shall be type 5 (see figure 2).
(3) Switch style shall be designated by the letters VLLS.
(4) There shall be no provisions for additional fuseholders.
(5) Wiring module basic circuitry shall be indicated by the designator 1 as specified to accomplish the switching actions specified in table I.
(6) Enclosed switch assemblies shall be indicated by a single letter (C).

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TABLE I. *Switch selection.*

Type designation						Application data						
Switch operation	Contact configuration	Switch style	Additional fuseholders	Wiring module	Enclosure	Switching action[1]	Stop position[1]	Dial position[2]	Handle position[3]	Wired contacts[4]	To ship connector[5]	Unwired contact[4]
M/R	5	VLLS	0	1	C	1,2,3,4 OFF	1&7			( )E OFF 5 ( )D OFF 4 ( )C LOAD 3 ( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	–[5]	( )F

Example: Type designation for the standard remote operated switch assembly shown with enclosure is R5VLLS0-1C; the manual version is M5VLLS0-1C.

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a style VLLS switch is considered to be that in which the flat on the handle shaft is up. The mark ∇ and the dial positions shown are used to designate the reference position.

[3]Normal position of a style VLLS switch is the position which coincides with normal circuit conditions in any application. Handle shall be adjustable to point upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification and table VI for typical switch wiring.

[5]See figures 4 and 5 and table VI for connector identifications and arrangements.

REQUIREMENTS:

Construction: Style RV5LLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 815-6189012.

Characteristics: Characteristics shown in table II shall apply to both styles of switch assemblies.

TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	2
Wired load switching poles	2
Number of positions	2 to 6
Break-before-make	Yes
Make-before-break	No

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TABLE II. *Characteristics – Continued.*

Characteristics	Value
Electrical rating and endurance loading:	
4 Amperes, 115 V, ac, resistive	20,000 Operations
3 Amperes, 115 V, ac, (0.3 p.f.), 60 Hz	20,000 Operations
3 Amperes, 115 V, ac, (0.1 p.f.), 400 Hz	20,000 Operations
4 Amperes, 28 V, dc, resistive	20,000 Operations
25 Microamperes, 1.5 Vdc	20,000 Operations
Remote operation:	
Transfer time:	
Minimum	0.030 second
Maximum	0.125 second
Actuating time:	
1 Position change	0.25 second
5 Position change (one direction)	1.50 seconds max.
Current drain:	
at 115 V, 3Ø, 60 Hz	1 Ampere rms per phase max. (0.70 Amp, typ.)
Contact resistance:	
At 4 amperes	0.025 ohm
At 25 microamperes	0.05 ohm
Total circuit resistance:	
At 4 amperes	0.025 ohm
At 25 microamperes	0.05 ohm
Torque – shaft rotation maximum – Initial	40 pound-inches
– After shock	40 pound-inches
– Stop strength	150 pound-inches
Maximum weight:	
Enclosed with wiring module	20 pounds
Without enclosure and wiring module	14 pounds
Attenuation	To be determined
Temperature and humidity (for rotational torque and endurance tests):	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) at 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

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Switch assembly modules: Style VLLS manually operated switch assemblies shall consist of the following modules as detailed on Drawing 815-6189012.

Switch assembly

Mechanism module
Switch contact module
Wiring module
Enclosure

Remote-operated, style VLLS switch assemblies shall have, in addition to the above,

Remote control module/Mechanism module (associated circuitry)
Control circuit module.

Switch function: Style VLLS switch assemblies shall provide two 6-position poles each consisting of 5 stationary contacts X, A, B, C, D, E, and F and a movable contact arranged as shown on figure 2. A stop shall prevent transfer of the movable contact into position X. Additional stops (movable) shall be provided which, when inserted as specified in table III, will limit movement to two through six consecutive positions. Stops shall be as detailed on Drawing 815-6189012 and shall be accessible from the front of the rotary description plate.

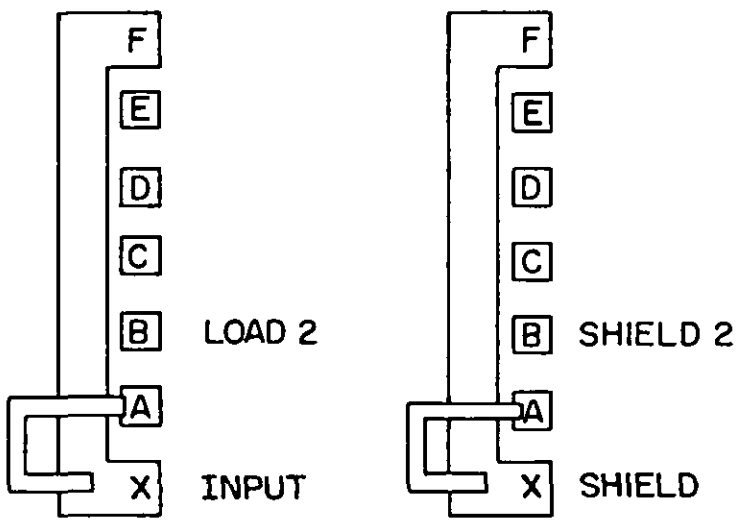


FIGURE 2. *Schematic typical operation of 1 pole.*

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TABLE III. *Stop positions versus switching action.*

Stop positions	Switch positions accessible	Switching action[1]
1 and 6	A-B-C-D-E	1-2-3-4-5
1 and 5	A-B-C-D	1-2-3-4-OFF
1 and 4	A-B-C	1-2-3-OFF or 1-2-3
1 and 3	A-B	1-2 or ON-OFF or OFF-ON

[1]With stops located as specified, a switch assembly shall provide from two to five active positions, any of which can be used as an OFF position by omitting external connections to the terminals in the desired OFF positions.

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter, A, B, C, D, or E shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact F is common with contact X in all poles. Contacts installed are signified by +, no contacts are signified by 0.

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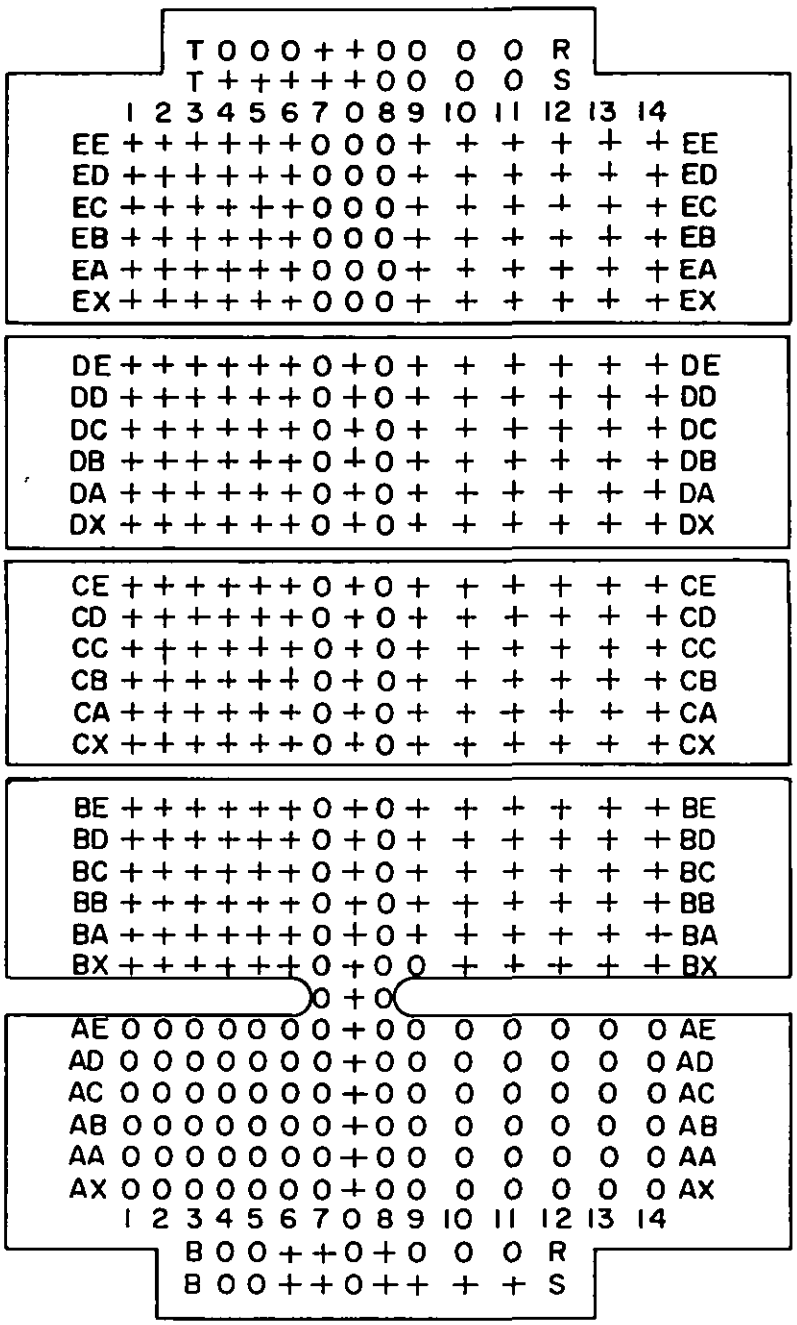
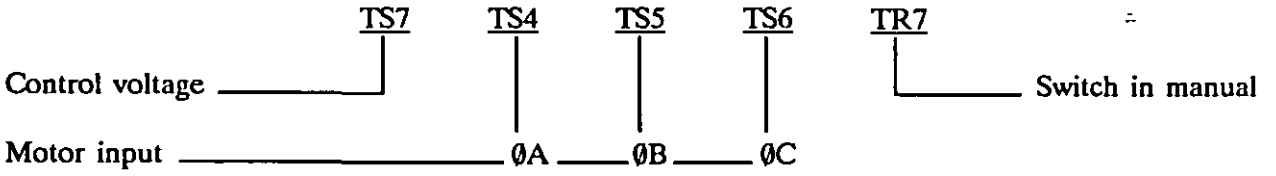


FIGURE 3. *Contact module marking, rear view.*

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Remote operation: Connections to the switch module for remote operation shall be made to contacts as follows (see table VII):



Remotely operated switch assembly shall move to the positions shown in table IV when the specified control voltage is applied.

TABLE IV. *Control voltage for remote operation.*

Control voltage	Closed switch contacts	XFMR tap[1,2]
0	X-A	A
16.4	X-B	B
32.8	X-C	C
49.2	X-D	D
65.6	X-E	E
82.0	X-F	F

[1]See figure 3 of MIL-S-24187 for typical test circuits for remote-operated switch assemblies.
[2]See figure 2 for typical pole positions corresponding to the transformer tap voltages A through F.

Table V lists the required connections in the wiring module for the remote control circuitry using type B-20 multi-strand grey wire of MIL-W-16878.

TABLE V. *Wiring connections for remote operations.*

Control wiring contact	Circuit jumper contact
BR7	AF0
BS7	AE0
BR8	AD0
BS8	AC0
BS9	AB0
BS10	AA0
BS11	AX0

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Enclosure and module:

Enclosure shall be furnished with the wiring module to connect the wiring switch contact module to the ship's wiring connector receptacles at the rear of the enclosure (see figure 5). The general arrangement of the wiring shall be as shown on figure 4. Interconnecting wiring shall be as shown in tables VI and VII and figures 7 and 8.

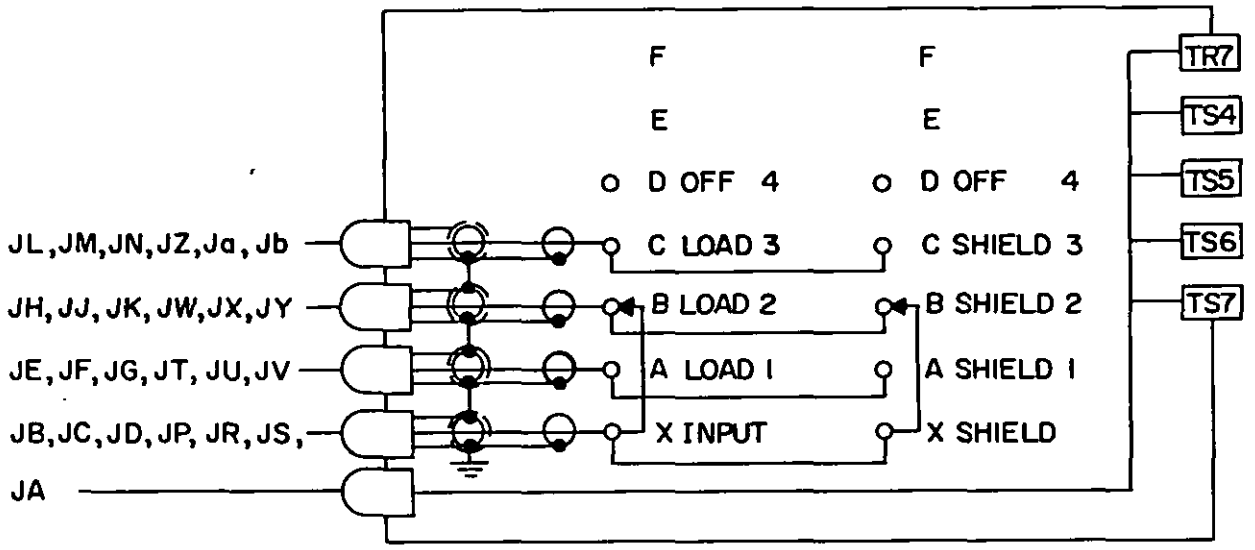


FIGURE 4. *Interconnecting wiring arrangement.*

Wiring module connections:

For switching circuits each coax connector will be wired to a specific contact of a switch contact module using the data shown in table VI and figure 7. Cable shall be type RG-174/U as specified in MIL-C-17/119. Schematic for 5VLLS on figure 8.

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TABLE VI. *Contacts in switch wiring module (typical).*

Switch wiring module contact			
Switch contacts X	Coax conductor column	Cable shield column	Wired to connector
E	2	1	JB
D	2	1	JE
C	2	1	JH
B	2	1	JL
E	4	3	JC
D	4	3	JF
C	4	3	JJ
B	4	3	JM

The 21-pin connector contacts shall be wired to contacts in the switch contact module as shown in table VII.

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TABLE VII. *Wiring module connections, 21-pin connector (JA).*

JA Contact	Switch contact	Switch service
1	TS4	PHASE A
2	TS5	PHASE B
3	TS6	PHASE C
6	TR7	SW in manual
4	TS7	Control
7	TR0	BFI Return
5	TS0	BFI Supply
8	CX0	
9	CA0	
10	CB0	Status
11	CC0	
12	CD0	Circuits
13	CE0	
14	DX0	
15	DA0	
16	DB0	
17	DC0	
18	DD0	
19	DE0	
20		Not wired
21		Not wired

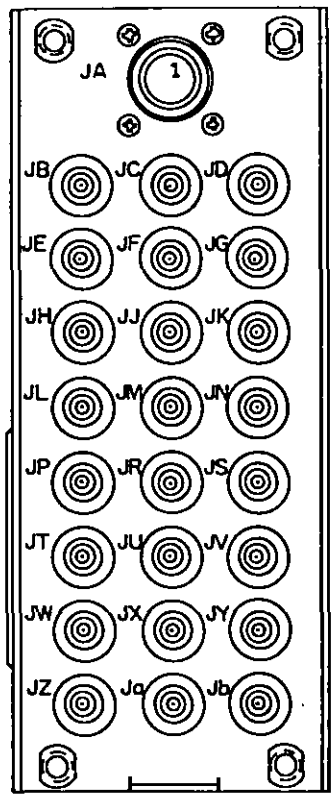


FIGURE 5. *Enclosed switch connector arrangements.*

JB-Jb connectors

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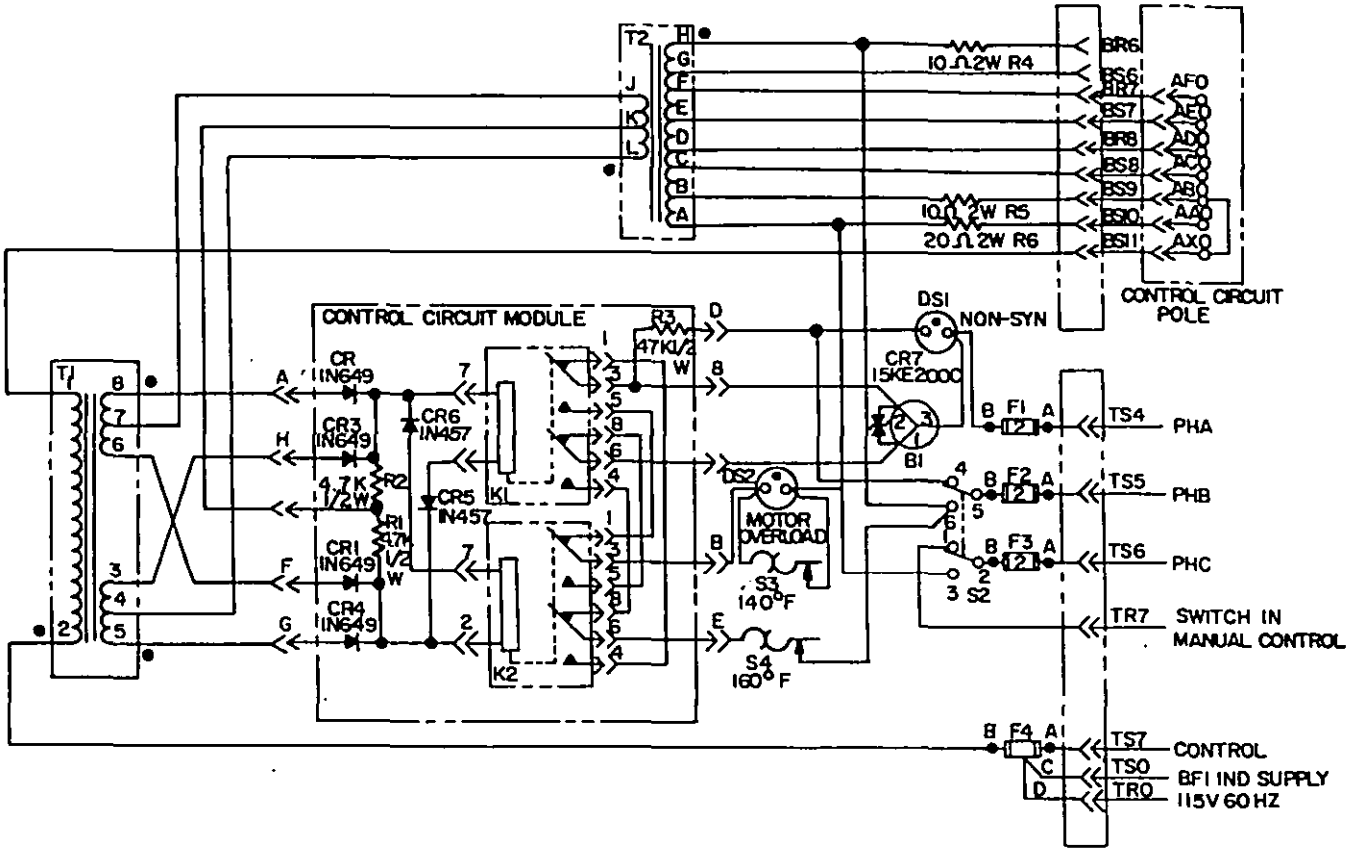
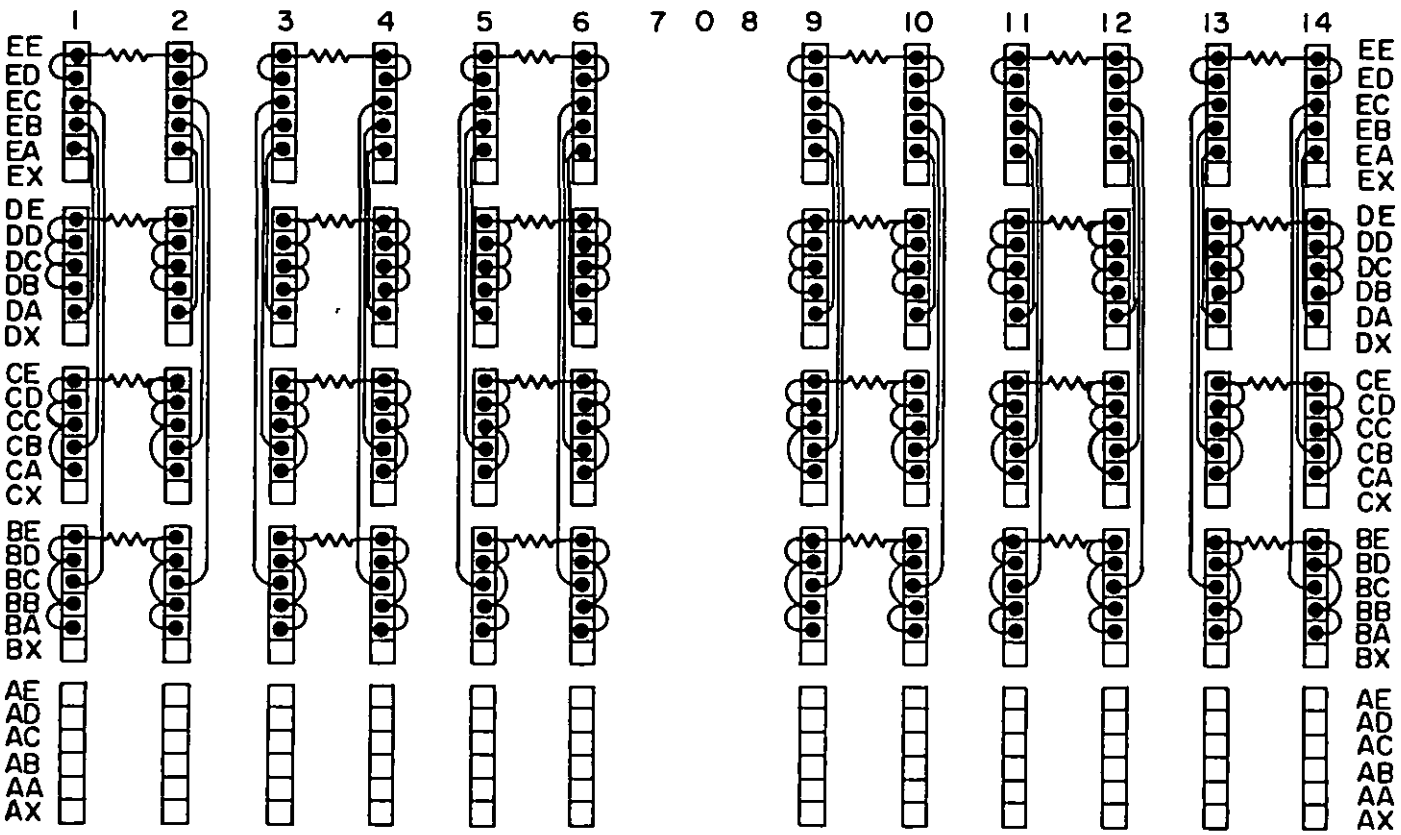


FIGURE 6. *Remote-control circuit.*

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1. Center conductors of coaxial cables are wired to contact X in columns 2, 4, 6, 10, 12 and 14. MIL-W-16878, type B-26, multi-strand white wire connections are in the same columns.
2. Shields of coaxial cables are wired to contact X in columns 1, 3, 5, 9, 11 and 13. MIL-W-16878, type B-26, multi-strand green wire connections are in the same columns.
3. Resistors R1 through R24 are 75 ohm, 1/4 W, 5 percent of MIL-R-39008/1.

FIGURE 7. *Interconnecting wiring.*

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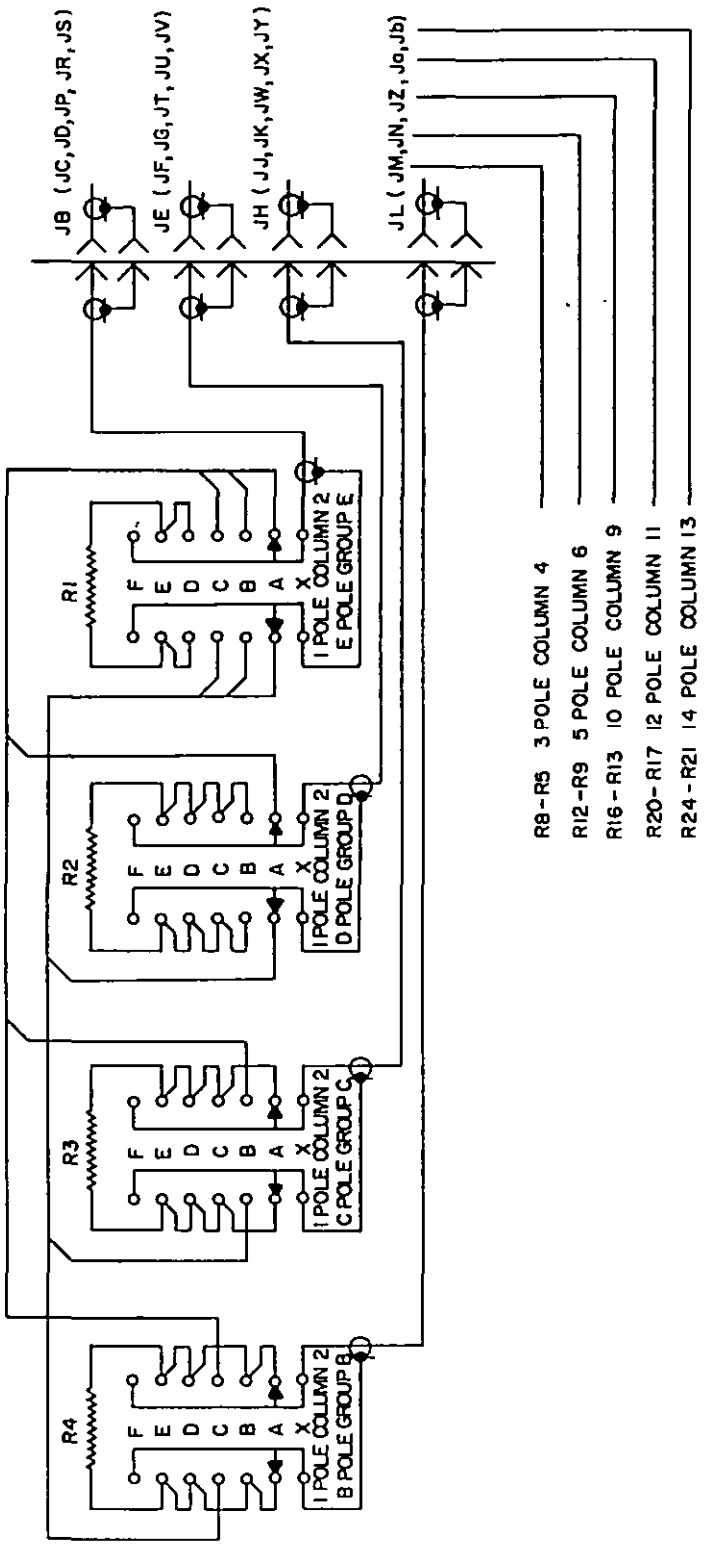


FIGURE 8. *5VLLS wiring module schematic.*

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Repair parts: Repair parts shall be as specified in table VIII.

TABLE VIII. *Repair parts.*

Drawing 815-6189012 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
2		Enclosed switch	1	1	1	1	2	2
	1	Enclosed switch	1	1	1	1	2	2
73	73	Stop screw	1	1	1	2	2	3
164	164	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer	2	2	2	4	6	8
	159	Spacer	2	2	2	4	6	8
	81	Fuseholder (less fuses)	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					

NOTES:
1. Refer to Drawing 803-4680149 for piece numbers.
2. One hundred percent repair lamps shall be provided.
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied. Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15.

Qualification samples: Two types R5VLLS0-1C switch assemblies shall be submitted for qualification inspection.

Manufacturers who have qualified the R5LLS0-1C are qualified for the R5VLLS0-1C.

Preparing activity:
Navy – SH
(Project 5930-N681-14)

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